

Sharp cardinal bounds for cup products on locally profinite spaces

Abstract

For every positive integer r , we construct a locally profinite Hausdorff space of cardinality and weight $\aleph_r$ with a degree-one class over $\mathbf{F}_2$ whose r -th cup power is nonzero. These sizes are minimal: nonzero degree- r sheaf cohomology, even with arbitrary abelian sheaf coefficients, forces both cardinality and weight at least $\aleph_r$. This answers Question 2.14 of Aoki. The construction uses a punctured involution quotient. Its cup-power nonvanishing is reduced to an elementary finite-difference argument: continuity gives finite exceptional sets, cardinal growth lets us avoid them uniformly, and symmetry makes induction possible. A separate argument with compact-open covers proves the universal lower bound. We give the full proof and a declaration-level guide to its Lean formalization.

1 The question and the answer

A locally profinite space has a basis of compact-open sets. Locally, such sets are particularly well behaved for sheaf cohomology: sections on a compact open have no higher derived functors. Nevertheless, assembling many compact opens can produce nonzero cohomology, and even nonzero products of positive-degree classes. How large must the space be for this to happen?

In [1, Theorem C], Aoki constructs, for each $n \geq 0$, a locally profinite space U of cardinality $\aleph_{2n+1}$ with classes $\eta_0, \dots, \eta_n \in H^1(U; \mathbf{F}_2)$ such that

$$\eta_0 \smile \dots \smile \eta_n \neq 0.$$

He then asks the following question.

Aoki's Question 2.14. Is Theorem C optimal in terms of the weight (or cardinality) of U ?

We prove that the sharp bound for a product of $n + 1$ degree-one classes is $\aleph_{n+1}$, for both cardinality and weight. All factors in the product can in fact be the same class. To keep the cohomological degree visible, we write $r = n + 1$ throughout the proof.

A *locally profinite Hausdorff space* will mean a locally compact, Hausdorff, totally disconnected space. A compact Hausdorff totally disconnected space is called *profinite*. Our examples are a profinite space with one point removed, so they also satisfy the convention used in [1]. The weight $w(V)$ is the least cardinality of a basis for the topology. All cohomology is ordinary sheaf cohomology; a constant coefficient such as $\mathbf{F}_2$ means the associated constant sheaf. Powers of a degree-one class are cup powers. We work with the axiom of choice; no continuum hypothesis is assumed.

Theorem 1.1. *For every integer $r \geq 1$, there exist a locally profinite Hausdorff space U_r and a class $\eta_r \in H^1(U_r; \mathbf{F}_2)$ such that*

$$|U_r| = w(U_r) = \aleph_r, \quad \eta_r^r \neq 0, \quad \eta_r^{r+1} = 0.$$

Moreover, for every abelian sheaf $\mathcal{M}$ on U_r ,

$$H^q(U_r; \mathcal{M}) = 0 \quad (q > r).$$

Conversely, for every locally profinite Hausdorff space V and every abelian sheaf $\mathcal{M}$ on V ,

$$H^r(V; \mathcal{M}) \neq 0 \implies |V| \geq \aleph_r \text{ and } w(V) \geq \aleph_r.$$

Thus $\aleph_r$ is both the least cardinality and the least weight of a locally profinite Hausdorff space supporting a nonzero product of r degree-one constant- $\mathbf{F}_2$ classes.^[L1]

The theorem shows that Aoki's bound is optimal when $n = 0$, and can be lowered for every $n \geq 1$. It also gives examples of cohomological dimension exactly r , where dimension is taken over all abelian sheaf coefficients.

1.1 Reading the paper and the formalization

The proof is accompanied by a Lean development. Small references such as^[L1] point to the source concordance in [appendix A](#), where each mathematical step is matched with explicit files, declarations, and line numbers. These references can be skipped on a first reading. The mathematical argument does not require knowledge of Lean. [Section 7](#) explains precisely what is checked and how to reproduce the verification.

1.2 The proof in five steps

There are two logically independent tasks: attaining the size $\aleph_r$, and excluding every smaller size. The first task contains the main construction. Its guiding idea is to build a cocycle for which being a boundary would force an impossible equation between functions.

1. **Put symmetry on coordinates of increasing sizes.** Take $r + 1$ coordinates with underlying sizes $\aleph_0, \dots, \aleph_r$. Each coordinate has a bit in $\mathbf{F}_2$ and a point at infinity. Identify points that differ by flipping every bit, and remove the all-infinite point. This is U_r . It has a cover by $r + 1$ opens, according to which coordinate is finite.
2. **Write the class and its power explicitly.** On the i -th open, choose the representative whose i -th bit is zero. On an overlap, the difference of the two choices is $z_{ij} = s_i + s_j$. These differences give η_r , and its top power is represented by

$$F_r = \prod_{j=0}^{r-1} (s_j + s_{j+1}).$$

The intersections in the cover are disjoint unions of compact opens. This makes their free sheaves projective, so an explicit finite resolution computes ordinary cohomology and its cup product.

3. **Extract the restrictions on a possible boundary.** If F_r were a boundary, it would be a sum $f_0 + \cdots + f_r$. Each f_i would be invariant under the simultaneous bit flip. It would also be constant outside finitely many points in its i -th coordinate, with the other coordinates fixed: this is exactly what continuity at infinity forces.
4. **Use the last coordinate to lower the degree.** There are only $\aleph_{r-1}$ configurations of the earlier coordinates. We can therefore choose one base point in the last coordinate that avoids every finite exceptional set for f_r . Adding the values at its two bits kills f_r and changes F_r into F_{r-1} . The other summands retain their symmetry and almost-constancy. Induction reaches a two-coordinate contradiction: an almost-constant function cannot differ by 1 from its bit flip everywhere.
5. **Bound the cohomology of every smaller space.** Independently, a cover by at most $\aleph_n$ compact opens forces cohomology to vanish above degree n . Countable covers can be disjointified; passing from one aleph to the next uses a continuous transfinite filtration and raises the projective-dimension bound by at most one. Both cardinality and weight control the size of a compact-open cover. This proves the lower bounds.

The roles of the three ingredients in the nonvanishing argument are worth keeping distinct. *Continuity* produces finite exceptional sets. *Cardinal growth* permits a choice avoiding all of them at once. *Symmetry* makes the two evaluations fit together into an induction. The projective resolution connects this elementary argument to sheaf cohomology.

1.3 The first example as a guide

For $r = 1$, the two coordinate sets have sizes $\aleph_0$ and $\aleph_1$, and the cocycle is simply $s_0 + s_1$. A hypothetical boundary would split it into two invariant functions, the first almost constant in coordinate zero and the second almost constant in coordinate one. Countably many finite exceptional sets in the second coordinate cannot exhaust its uncountable set of base points. Evaluate at both bits of a base point outside their union. The second function cancels, and the resulting slice of the first would have to change by 1 under bit flip. Far enough from its own finite exceptional set, however, it is constant. This contradiction is the base case for every higher cup power.

We now carry out the construction in [section 2](#), the cohomological calculation in [sections 3 and 4](#), and the nonvanishing argument in [section 5](#). The independent optimality proof occupies [section 6](#).

2 The space and its compact-open pieces

Fix $r \geq 1$. Choose discrete sets A_i with $|A_i| = \aleph_i$, for $0 \leq i \leq r$, and set

$$D_i = A_i \times \mathbf{F}_2, \quad K_i = D_i \sqcup \{\infty_i\}, \quad K = \prod_{i=0}^r K_i.$$

Give K_i the one-point compactification topology: every point of D_i is isolated, and the neighborhoods of ∞_i contain all but finitely many points of D_i . The spaces K_i and their finite product K are profinite.

Let τ flip the bit in every finite coordinate and fix each point at infinity:

$$\tau(a, s) = (a, s + 1), \quad \tau(\infty_i) = \infty_i.$$

This defines a continuous involution of K . Its only fixed point is $p = (\infty_0, \dots, \infty_r)$, because any finite coordinate changes its bit. Write

$$q : K \longrightarrow \bar{K} = K/\langle \tau \rangle, \quad U_r = \bar{K} \setminus \{q(p)\}.$$

Taking the compact quotient first makes the topology of the example particularly transparent. The quotient is compact Hausdorff: its orbit relation is the union of the diagonal and the graph of a homeomorphism, hence is closed. It is also zero-dimensional. Indeed, in the inverse image of any open neighborhood of an orbit, choose a clopen neighborhood of one representative; its union with its translate is invariant and clopen, and descends to a clopen neighborhood in the quotient. The orbit map is open, since the saturation of an open set O is $O \cup \tau(O)$. Thus $\bar{K}$ is profinite and U_r is locally profinite Hausdorff.^[L2]

2.1 Choosing a representative on each chart

Let $W_i \subseteq U_r$ consist of the orbits whose i -th coordinate is finite. These opens cover U_r : the only orbit with no finite coordinate was removed. On W_i , each orbit has exactly one representative whose i -th bit is zero. This representative depends continuously on the orbit, because the bit-zero slice is open and the quotient map restricts to an open continuous bijection from that slice onto W_i . We call this the *gauge choice* on W_i .^[L3]

Lemma 2.1. *For every nonempty $I \subseteq \{0, \dots, r\}$, the intersection $W_I = \bigcap_{i \in I} W_i$ is a disjoint union of compact-open subsets of U_r .*

Proof. Choose an anchor $a \in I$ and use its gauge choice. Prescribe every finite coordinate $(a_i, s_i) \in D_i$ for $i \in I$, with $s_a = 0$, and leave all coordinates outside I unrestricted. This gives a clopen cylinder in K . Its image in the quotient is compact and open, and avoids $q(p)$.

These images are disjoint for distinct prescriptions. If two meet, the corresponding representatives either agree or are exchanged by τ . Exchange is impossible because both anchor bits are zero; agreement forces identical prescriptions. Every orbit in W_I has one such prescription, so the images cover W_I . $\square$

These compact-open pieces explain why the cover will compute cohomology. The intersections need not themselves be compact; it is their decomposition into disjoint compact pieces that matters.

2.2 Computing cardinality and weight

Proposition 2.2. *The space U_r satisfies $|U_r| = w(U_r) = \aleph_r$.*

Proof. The finite product K has cardinality $\aleph_r$, the size of its largest factor. A quotient and then a subspace cannot increase cardinality, so $|U_r| \leq \aleph_r$.

For the weight estimate, recall that an infinite compact Hausdorff space T has $w(T) \leq |T|$. Here is a useful direct proof. For each ordered pair of distinct points x, y , choose disjoint neighborhoods O_{xy} of x and V_{xy} of y . Given an open neighborhood N of x , finitely many of the V_{xy} cover the compact set $T \setminus N$. The intersection of the corresponding O_{xy} is contained in N . These finite intersections, for all x , form a basis with at most $|T|$ elements.

An open quotient map does not increase weight, because the images of basis elements form a basis. Subspaces do not increase weight either. Consequently,

$$w(U_r) \leq w(\bar{K}) \leq w(K) \leq \aleph_r.$$

For both reverse inequalities, consider the axis whose only finite coordinate is the last one. Every orbit there has a unique representative with last coordinate $(a, 0)$, so the axis is a copy of A_r . It is a discrete subspace: the open condition that the last base coordinate equal a isolates its corresponding axis point. We obtain $\aleph_r$ distinct points, and any basis for U_r must contain distinct neighborhoods isolating these points in the axis. Thus both cardinality and weight are at least $\aleph_r$.^[L4] $\square$

3 A finite resolution that computes cohomology

The geometry above serves two purposes. It bounds the size of the space, and it provides a cover with projective free sheaves on all its intersections. We spell out the latter point so that the passage from functions to ordinary sheaf cohomology is explicit.

3.1 Free sheaves on opens

For an open W in a space X , let $P(W)$ be the free abelian sheaf represented by W . Concretely, sheafify the free abelian presheaf on the representable presheaf of W . What we use is its natural representing property:

$$\mathrm{Hom}(P(W), \mathcal{M}) \cong \Gamma(W, \mathcal{M}).$$

An inclusion $V \subseteq W$ induces $P(V) \rightarrow P(W)$, and $P(X) \cong \underline{\mathbb{Z}}_X$. For a commutative ring R , write $P_R(W)$ for the corresponding free sheaf of R -modules. It represents sections in that category, with $P_R(X) \cong \underline{R}_X$.

Lemma 3.1. *Suppose X is Hausdorff and totally disconnected. If W is a disjoint union of compact-open subsets of X , then $P(W)$ and $P_R(W)$ are projective.*^[L5]

Proof. To prove projectivity, it suffices to lift a section over W through an epimorphism of sheaves. Epimorphisms are locally surjective, so local lifts exist. On a compact open C , choose finitely many clopen sets refining a local lifting cover. This is possible because C is compact Hausdorff and totally disconnected, hence zero-dimensional. Subtract earlier clopen sets from later ones to make the finite cover disjoint. The local lifts then glue, with no compatibility conditions on overlaps, to a section on C . Do this on each compact-open component of W and glue again. The representing property gives projectivity in both coefficient categories. $\square$

3.2 The ordered complex and its exactness

For the cover $W_0, \dots, W_r$, set

$$P_k = \bigoplus_{0 \leq i_0 < \dots < i_k \leq r} P(W_{i_0} \cap \dots \cap W_{i_k}).$$

Deleting one index enlarges the intersection and induces a map of free sheaves. Let the differential be the alternating sum of these maps. The augmentation $P_0 \rightarrow \underline{\mathbb{Z}}_{U_r}$ is induced by the inclusions of the charts. The usual face-deletion identities imply that the differential squares to zero.

Proposition 3.2. *The augmented complex*

$$0 \longrightarrow P_r \longrightarrow \cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow \underline{\mathbf{Z}}_{U_r} \longrightarrow 0$$

is a projective resolution. Replacing each free integral sheaf by its R -module counterpart gives a projective resolution $P_\bullet^R \rightarrow \underline{R}_{U_r}$.^[L6]

Proof. All terms are projective by [lemmas 2.1](#) and [3.1](#). For exactness, take a stalk at $x \in U_r$. The stalk of $P(W)$ is zero when $x \notin W$; if $x \in V \subseteq W$, the map $P(V)_x \rightarrow P(W)_x$ is an isomorphism. These assertions follow from the representing property, or from the stalk–skyscraper adjunction.

Let a be the least index with $x \in W_a$. The surviving faces form the simplex of charts containing x , and its augmented chain complex contracts to the vertex a . Explicitly, prepend a to an increasing face whose first vertex is larger than a , using the inverse of the stalk inclusion; on a face starting at a , set the contraction to zero. Faces starting below a have zero stalk. In the alternating boundary formula, deleting the inserted vertex gives the original face and all other terms cancel in pairs. Thus $dh + hd = 1$, with the augmentation section included in degree zero. The augmented stalk complex is exact. Exactness of sheaves is detected on stalks. The same contraction works for the module sheaves. $\square$

Applying $\text{Hom}(-, \mathcal{M})$ now computes ordinary sheaf cohomology, since

$$H^q(U_r; \mathcal{M}) = \text{Ext}_{\text{Sh}(U_r; \mathbf{Ab})}^q(\underline{\mathbf{Z}}, \mathcal{M}).$$

This also identifies the cochain complex explicitly: its degree- k term is the product of the section groups on intersections of $k + 1$ charts, and its differential is the alternating sum of restrictions. In particular, with constant $\mathbf{F}_2$ coefficients, these sections are locally constant $\mathbf{F}_2$ -valued functions. Because there are no faces of degree greater than r , we have proved

$$H^q(U_r; \mathcal{M}) = 0 \quad (q > r)$$

for every abelian coefficient sheaf.^[L7]

The remaining question is whether a particular cochain in the last possible degree is a boundary. We first identify that cochain as a power of a degree-one class.

4 The class and the cup-product formula

4.1 Transition functions

On $W_i \cap W_j$, define

$$z_{ij} = s_i + s_j \in \mathbf{F}_2,$$

using any representative of the orbit. This is well defined: flipping both bits adds $1 + 1 = 0$. It is locally constant, as is visible on the compact-open cells. For three indices,

$$z_{ij} + z_{jk} = z_{ik}.$$

Thus $z = (z_{ij})_{i < j}$ is a degree-one cocycle in the complex of [section 3](#). Let

$$\eta_r = [z] \in H^1(U_r; \mathbf{F}_2).$$

Geometrically, z_{ij} records whether the gauge choices on the two charts agree or differ by the involution. This observation motivates the cocycle; the proof uses the displayed functions and their identities directly.^[L8]

4.2 Why multiplication is computed by adjacent differences

Proposition 4.1. *For $1 \leq k \leq r$, the class η_r^k is represented on the face $[i_0, \dots, i_k]$ by*

$$z_{i_0 i_1} z_{i_1 i_2} \cdots z_{i_{k-1} i_k}.$$

In particular, on the full intersection and pulled back to configurations in which every coordinate is finite, η_r^r is represented by

$$F_r((a_i, s_i)_{i=0}^r) = \prod_{j=0}^{r-1} (s_j + s_{j+1}). \quad (1)$$

The familiar ordered Čech cup formula suggests this answer. We give a chain-level proof, which also fixes its relation to ordinary sheaf cohomology and to the operation used in the formalization.

Proof. Use the resolution $P_{\bullet}^{\mathbf{F}_2}$ in sheaves of $\mathbf{F}_2$ -modules. Its graded Ext group

$$\mathrm{Ext}_{\mathrm{Sh}(U_r; \mathbf{F}_2\text{-Mod})}^*(\underline{\mathbf{F}_2}, \underline{\mathbf{F}_2})$$

has the Yoneda product: multiplication is composition of the corresponding morphisms in the derived category. This is the standard realization of the cup algebra of derived global sections.

We compare this description with the ordinary cohomology already computed. The integral and module resolutions are each independently projective and exact. Their Hom complexes, with constant $\mathbf{F}_2$ as target, identify with the same complex of sections on chart intersections. The identifications commute with restrictions and hence with differentials. At degree zero the rank-one generator maps to the constant function 1. This section-induced comparison identifies the module Ext description with ordinary $H^*(U_r; \mathbf{F}_2)$, with its usual cup multiplication.^[L9]

For a direct product calculation, write $d_k : P_{k+1}^{\mathbf{F}_2} \rightarrow P_k^{\mathbf{F}_2}$, and define

$$L_k : P_{k+1}^{\mathbf{F}_2} \longrightarrow P_k^{\mathbf{F}_2}$$

on a face $[i_0, \dots, i_{k+1}]$ as follows: multiply by $z_{i_0 i_1}$, delete the first vertex, and map into the summand indexed by $[i_1, \dots, i_{k+1}]$. A locally constant scalar acts on sheaf sections by gluing the constant scalar actions on its open fibers, so this construction defines a sheaf morphism.

The cocycle identity gives the chain-lift relation

$$L_k \circ d_{k+1} = -d_k \circ L_{k+1}.$$

Indeed, terms deleting a vertex after the first two cancel by the face identities. The remaining terms combine by $z_{i_1 i_2} + z_{i_0 i_2} = z_{i_0 i_1}$. In characteristic two the minus sign is the same as a plus sign. If $\epsilon : P_0^{\mathbf{F}_2} \rightarrow \underline{\mathbf{F}_2}$ is the augmentation, then $\epsilon \circ L_0$, under the section comparison, is precisely z . Thus L lifts the degree-one cocycle to a shifted map of the resolution.

Set $c_0 = \epsilon$ and $c_{k+1} = c_k \circ L_k$. The chain identity shows that these are cocycles. To see which classes they represent, recall that the augmentation is inverted when passing from the resolution to its resolved object in the derived category. In composing two morphisms represented through the resolution, the middle augmentation and its inverse cancel. Consequently composing with the lift L computes multiplication by η_r . Induction shows that c_k represents η_r^k for $k \geq 1$.

Finally, each application of L removes the current first vertex and multiplies by its transition to the next one. On a face of length $k + 1$, the composite therefore multiplies successively by $z_{i_0 i_1}, z_{i_1 i_2}, \dots, z_{i_{k-1} i_k}$. This is the claimed formula. ^[L10] $\square$

The product formula gives a useful simplification: although the space has points at infinity, the top intersection has none. Its pulled-back cochain F_r is just a function on a product of discrete sets. The points at infinity return only when we ask whether this function is a sum of restrictions of continuous functions on the intersections one degree lower. That is where the nonvanishing argument begins.

5 Why the top cup power does not vanish

The cup-product calculation has reduced nonvanishing to a question about functions. We must show that the product in (1) cannot be the restriction of a coboundary. Two features of the construction work together: approaching infinity forces eventual constancy in a coordinate, and the strictly increasing coordinate cardinalities let us use this constancy simultaneously for every choice of the other coordinates.

Recall that $D_i = A_i \times \mathbf{F}_2$, with $|A_i| = \aleph_i$. Write

$$C_r = \prod_{i=0}^r D_i, \quad \sigma_r((a_i, s_i)_{i=0}^r) = (a_i, s_i + 1)_{i=0}^r.$$

Thus C_r consists of the all-finite configurations, before passing to the quotient. The pullback of the top product cochain is

$$F_r((a_i, s_i)_{i=0}^r) = \prod_{j=0}^{r-1} (s_j + s_{j+1}).$$

We use the empty-product convention $F_0 = 1$. Throughout this section, all function values and sums are in $\mathbf{F}_2$.

The function F_r depends only on the bits, not on the base points a_i . For instance, it takes the value one whenever consecutive bits are different. This observation alone says nothing about cohomological nonvanishing: a nonzero function can still be a coboundary. The role of the large base sets is to restrict which invariant functions can occur in a boundary, through the continuity requirements at the points at infinity. We exploit those requirements without trying to describe all possible continuous functions on the intersections.

5.1 What a boundary would imply

Call a function $f : C_r \rightarrow \mathbf{F}_2$ *almost constant in coordinate i* if, after fixing every other coordinate, its restriction to D_i is constant outside a finite set. Neither the exceptional set nor the eventual constant is required to be independent of the fixed coordinates.

Lemma 5.1. *Suppose $r \geq 1$. If the top product cochain is a coboundary, then*

$$F_r = \sum_{i=0}^r f_i,$$

where every $f_i : C_r \rightarrow \mathbf{F}_2$ is invariant under σ_r and almost constant in coordinate i .

Proof. An ordered cochain one degree below the top has a component on each intersection

$$W_{\hat{i}} = \bigcap_{j \neq i} W_j.$$

These components are locally constant $\mathbf{F}_2$ -valued functions. The top differential is their sum after restriction to the full intersection: its alternating signs disappear in characteristic two. Pulling the components back to C_r gives the functions f_i . Because they are pulled back from the orbit space, they are invariant under the simultaneous flip.

Fix every coordinate except i . Allowing coordinate i to vary through its entire one-point compactification defines a continuous map

$$D_i^+ \longrightarrow W_{\hat{i}}.$$

This map is defined even at infinity. Indeed, $r \geq 1$, so at least one other coordinate remains finite; the removed all-infinite orbit is never encountered. Compose with the component function on $W_{\hat{i}}$. Since $\mathbf{F}_2$ is discrete, continuity at infinity says that this composite equals its value at infinity on a cofinite subset of D_i . This is exactly the required almost constancy. Notice that this concerns the whole doubled set $D_i = A_i \times \mathbf{F}_2$. Outside a finite set, both bit values therefore have the same eventual value. Two unrelated eventual values on the separate bit-zero and bit-one subsets would not suffice for the argument below. $\square$

Only this necessary condition is used. In particular, we do not claim that an arbitrary family of coordinatewise almost-constant functions extends to jointly continuous functions on the deleted intersections.^[L11]

5.2 One choice avoiding all exceptional sets

The exceptional set just obtained can depend on many other coordinates. The following elementary counting argument supplies a single choice that avoids all of them.

Lemma 5.2 (Uniform thinning). *Let P be infinite and $|P| < |A|$. For each $y \in P$, let $N_y \subseteq A \times \mathbf{F}_2$ be finite. There is an $a \in A$ such that both $(a, 0)$ and $(a, 1)$ lie outside every N_y .*

Proof. Let $\pi : A \times \mathbf{F}_2 \rightarrow A$ be the first-coordinate projection. Since each $\pi(N_y)$ is finite,

$$\left| \bigcup_{y \in P} \pi(N_y) \right| \leq |P| \cdot \aleph_0 = |P| < |A|.$$

Choose a outside this union. No point of either form $(a, 0)$ or $(a, 1)$ can belong to any of the exceptional sets. $\square$

For the application, the index set is C_{r-1} , whose cardinality is $\aleph_{r-1}$, while the last base set has cardinality $\aleph_r$. This strict inequality is the reason for increasing the coordinate sizes. The argument uses no uniform bound on the sizes of the finite exceptional sets, and no common eventual value. It is simply a bound on a union of finite sets, so no regularity assumption on the larger cardinal is being used.

Choosing a point separately for each configuration would be much weaker. It would give a base point depending on the remaining coordinates, and then taking the associated slices could destroy almost constancy in those coordinates. The uniform choice instead gives two fixed slices of every function. Their algebraic and continuity properties can then be compared directly. This is the only place where the strict inequality between successive coordinate sizes is used; the remaining reduction is elementary linear algebra.^[L12]

5.3 Removing one coordinate by a finite difference

Proposition 5.3. *For every $r \geq 1$, there is no decomposition*

$$F_r = \sum_{i=0}^r f_i$$

in which f_i is invariant under σ_r and almost constant in coordinate i .

Proof. We establish a reduction in degree and then prove the two-coordinate base case separately. Suppose that such a decomposition has been given. For each $y \in C_{r-1}$, choose an eventual value c_y and a finite exceptional set $N_y \subseteq D_r$ for the last-coordinate function $z \mapsto f_r(y, z)$. Lemma 5.2 supplies $a \in A_r$ avoiding all these exceptions.

For any function $f : C_r \rightarrow \mathbf{F}_2$, define its two-bit difference by

$$(\Delta_a f)(y) = f(y, (a, 0)) + f(y, (a, 1)), \quad y \in C_{r-1}.$$

For our chosen a , both values of the last summand equal c_y . Consequently, $\Delta_a f_r = 0$. In contrast, the product function loses exactly its last factor:

$$\begin{aligned} (\Delta_a F_r)(y) &= F_{r-1}(y)(s_{r-1} + (s_{r-1} + 1)) \\ &= F_{r-1}(y). \end{aligned}$$

For $i < r$, the function $\Delta_a f_i$ remains almost constant in coordinate i . Fix the other coordinates and take the union of the finite exceptional sets for its two slices. Outside this union, the sum of their two eventual values is constant.

The difference also preserves invariance. If f is invariant under the simultaneous flip, then

$$\begin{aligned} (\Delta_a f)(\sigma_{r-1} y) &= f(\sigma_{r-1} y, (a, 0)) + f(\sigma_{r-1} y, (a, 1)) \\ &= f(y, (a, 1)) + f(y, (a, 0)) \\ &= (\Delta_a f)(y). \end{aligned}$$

Here it matters that the two points have the same base coordinate: the flip exchanges the two summands. Thus, for $r \geq 2$, applying Δ_a gives

$$F_{r-1} = \sum_{i=0}^{r-1} \Delta_a f_i,$$

with all the required properties in one lower positive degree. This contradicts the induction hypothesis once the case $r = 1$ is known.

In this reduction, the largest coordinate and the summand controlled by that coordinate disappear together. Each surviving summand retains its own almost-constancy condition, while the target function becomes the same adjacent-bit product on the shorter configuration space. The operation sums only two values, so no infinite sums or convergence questions enter the reduction.

For that base case, the reduction gives $\Delta_a f_0 = 1$. Retain the single slice

$$g(x) = f_0(x, (a, 0)), \quad x \in D_0.$$

It is almost constant because f_0 has that property in coordinate zero. Invariance of f_0 , applied to $(x, (a, 1))$, gives $g(\sigma_0 x) = f_0(x, (a, 1))$. Hence

$$g(x) + g(\sigma_0 x) = 1 \quad \text{for every } x \in D_0.$$

Choose a finite exceptional set $N \subseteq D_0$ for g , with eventual value c . The set A_0 is infinite, so it contains a_0 outside the first-coordinate projection of N . Both $(a_0, 0)$ and $(a_0, 1)$ then lie outside N . Substituting the first of these points into the displayed identity gives $c + c = 1$, contradicting $c + c = 0$. This proves the base case and the induction.^[L13] $\square$

Remark 5.4. Two tempting shortcuts would lose the argument. First, the single slice g need not be invariant; the proof uses the relation between g and its flipped value instead. Second, one cannot continue a bare nondecomposition induction down to degree zero. The function $F_0 = 1$ is itself invariant and almost constant, so the corresponding nondecomposition assertion is false. The base case succeeds because it retains the extra information that this constant function is the flip difference of an almost-constant slice.

5.4 Returning to cohomology

The top cochain in (1) represents η_r^r , by the chain-lift calculation. If this class were zero, the projective resolution would express that cochain as a coboundary. Lemma 5.1 would then produce the decomposition forbidden by Proposition 5.3. We conclude that

$$\eta_r^r \neq 0.$$

The same resolution has length r , so it also gives $H^{r+1}(U_r; \mathbf{F}_2) = 0$, and therefore $\eta_r^{r+1} = 0$. Combined with the cardinality and weight of U_r , these statements prove the existence assertions of Theorem 1.1.^[L14]

6 The universal lower bound

The examples constructed above attain the least possible cardinality and weight. We prove this by an estimate for arbitrary abelian coefficient sheaves, independent of the quotient construction. The relevant invariant is the number of compact opens needed to cover a space.

Recall that $P(W)$ denotes the integral free sheaf represented by an open subset $W \subseteq X$, so that

$$\mathrm{Hom}(P(W), \mathcal{M}) \cong \Gamma(W, \mathcal{M}).$$

In particular, $P(X) \cong \underline{\mathbb{Z}}_X$. Throughout this section, projective dimension is understood in its Ext-vanishing sense: $\text{pd}(P) \leq n$ means that $\text{Ext}^d(P, M) = 0$ for every coefficient object M and every $d > n$. The category of abelian sheaves has enough injectives, so dimension shifting in the coefficient variable is available. The indexing begins with countable covers: they already give projective free-open sheaves, rather than merely a dimension bound of one. Each passage to the next infinite cardinal costs at most one further degree. Keeping this starting point explicit will give precisely the lower bound $\aleph_r$ in degree r .

6.1 Continuous transfinite extensions

The homological input is a closure property whose proof is worth recording. Continuity of the filtration is essential to its use below.

Lemma 6.1. *Let $\mathcal{A}$ be an abelian category with enough injectives. Suppose $(P_\alpha)_{\alpha \leq \lambda}$ is a well-ordered filtration with $P_0 = 0$, monomorphic successor maps, and*

$$P_\delta \cong \text{colim}_{\alpha < \delta} P_\alpha$$

at every nonzero limit ordinal $\delta \leq \lambda$. If each successive quotient $Q_\alpha = P_{\alpha+1}/P_\alpha$ has projective dimension at most n , then $\text{pd}(P_\lambda) \leq n$.

Proof. First fix $Y \in \mathcal{A}$ and assume only that $\text{Ext}^1(Q_\alpha, Y) = 0$ for all $\alpha < \lambda$. Choose an injective presentation

$$0 \longrightarrow Y \longrightarrow I \xrightarrow{p} C \longrightarrow 0.$$

The long exact Ext sequence shows that $\text{Ext}^1(Q, Y) = 0$ precisely when every map $Q \rightarrow C$ lifts through p . We will prove this lifting property for P_λ .

Given $f : P_\lambda \rightarrow C$, denote its restriction to P_α by f_α . Construct compatible maps $g_\alpha : P_\alpha \rightarrow I$ satisfying $pg_\alpha = f_\alpha$. There is a unique choice at stage zero. At a successor stage, injectivity of I extends g_α to a map $h : P_{\alpha+1} \rightarrow I$. This extension may not lift $f_{\alpha+1}$, but the discrepancy $f_{\alpha+1} - ph$ vanishes on P_α . Writing $\pi_\alpha : P_{\alpha+1} \rightarrow Q_\alpha$ for the quotient map, there is therefore a map $b : Q_\alpha \rightarrow C$ such that

$$f_{\alpha+1} - ph = b\pi_\alpha.$$

By the assumed Ext vanishing, choose $\tilde{b} : Q_\alpha \rightarrow I$ with $p\tilde{b} = b$. Then $g_{\alpha+1} = h + \tilde{b}\pi_\alpha$ is both an extension of g_α and a lift of $f_{\alpha+1}$. At a limit stage, the compatible earlier lifts induce a map from the colimit. Their identities with the restrictions of f persist by the colimit's universal property. This constructs g_λ , proving $\text{Ext}^1(P_\lambda, Y) = 0$.

Now fix any $d > n$ and any coefficient object Y . Choose successive injective presentations

$$0 \longrightarrow Y_j \longrightarrow I_j \longrightarrow Y_{j+1} \longrightarrow 0, \quad Y_0 = Y, \quad 0 \leq j < d-1.$$

Dimension shifting gives, for every object Q ,

$$\text{Ext}^d(Q, Y) \cong \text{Ext}^1(Q, Y_{d-1}).$$

The same coefficient object Y_{d-1} works for all successive quotients. Their dimension bounds make the right-hand side vanish, so the degree-one argument applies. Shifting back gives $\text{Ext}^d(P_\lambda, Y) = 0$. Since $d > n$ and Y were arbitrary, the required projective-dimension bound follows.^[L15] $\square$

Remark 6.2. The proof constructs compatible lifts; it does not commute Ext with an arbitrary filtered colimit. In particular, the closure statement requires no separate assumption that arbitrary filtered colimits are exact.

6.2 Counting compact opens

Proposition 6.3. *Let X be Hausdorff and totally disconnected. If an open subset $W \subseteq X$ is a union of at most $\aleph_n$ compact-open subsets of X , then $\text{pd}(P(W)) \leq n$.*

Proof. We induct on n . For $n = 0$, enumerate the given compact opens as $K_0, K_1, \dots$, allowing empty terms, and set

$$L_j = K_j \setminus \bigcup_{i < j} K_i.$$

The earlier union is finite and compact, hence closed in the Hausdorff space X ; consequently L_j is open. It is also closed in the compact set K_j , since the earlier union is open. Thus the L_j are compact open, pairwise disjoint, and have union W . Lemma 3.1 makes $P(W)$ projective. This includes finite and empty covers. Concretely, sections through a sheaf epimorphism lift separately on every L_j , and the lifts glue without compatibility conditions because the pieces are disjoint. This explains why no finiteness condition on the disjoint family is needed. Local compactness of the ambient space is not required for this step or for the proposition.

Assume the assertion for n . Pad a family of at most $\aleph_{n+1}$ compact opens with empty sets to index it by the initial ordinal $\kappa = \omega_{n+1}$, and put

$$W_\alpha = \bigcup_{\beta < \alpha} K_\beta \quad (\alpha \leq \kappa), \quad W_\kappa = W.$$

These opens give a continuous filtration of free sheaves. Indeed, inclusions induce monomorphisms: on stalks the free-open sheaf is $\mathbf{Z}$ inside its representing open and zero outside, and the nonzero inclusion maps are identities. Moreover, the free sheaf of a directed union is the colimit of the free sheaves of its members. By the representing property, this assertion says exactly that compatible sections on the members glue uniquely to a section on their union. Hence $P(W_0) = 0$, continuity holds at limit ordinals, and the final object is $P(W)$.

For two opens V, T , the free-open Mayer–Vietoris sequence is

$$0 \longrightarrow P(V \cap T) \longrightarrow P(V) \oplus P(T) \longrightarrow P(V \cup T) \longrightarrow 0.$$

The first map is the signed pair of inclusions, and the second adds the inclusions into the union. Exactness can be checked on stalks; on the intersection the maps are $u \mapsto (u, -u)$ and $(v, t) \mapsto v + t$. It follows, by taking the quotient by $P(V)$, that the successive quotient $Q_\alpha = P(W_{\alpha+1})/P(W_\alpha)$ fits into

$$0 \longrightarrow P(W_\alpha \cap K_\alpha) \longrightarrow P(K_\alpha) \longrightarrow Q_\alpha \longrightarrow 0.$$

For every $\alpha < \kappa$, the initial-ordinal property gives $|\alpha| < \aleph_{n+1}$, hence $|\alpha| \leq \aleph_n$. Since

$$W_\alpha \cap K_\alpha = \bigcup_{\beta < \alpha} (K_\beta \cap K_\alpha),$$

the induction hypothesis bounds the left-hand term's projective dimension by n . The middle term is projective by Lemma 3.1. For $d > n + 1$, the long exact Ext sequence identifies

$\text{Ext}^d(Q_\alpha, M)$ with $\text{Ext}^{d-1}(P(W_\alpha \cap K_\alpha), M)$, which vanishes. Thus $\text{pd}(Q_\alpha) \leq n + 1$. Lemma 6.1 completes the induction. Only the size of each proper initial segment of ω_{n+1} enters the argument. There is no appeal to the continuum hypothesis or to any extra cardinal-arithmetic assumption.^[L16] $\square$

Taking $W = X$, the identification $P(X) \cong \underline{\mathbb{Z}}_X$ immediately gives the following consequence.

Corollary 6.4. *If a Hausdorff totally disconnected space X has a compact-open cover of cardinality at most $\aleph_n$, then $H^d(X; \mathcal{M}) = 0$ for every abelian sheaf $\mathcal{M}$ and every $d > n$.*

6.3 Cardinality and weight

Corollary 6.5. *Let V be locally compact, Hausdorff, and totally disconnected, let $r \geq 1$, and let $\mathcal{M}$ be any abelian sheaf on V . Then*

$$H^r(V; \mathcal{M}) \neq 0 \implies |V| \geq \aleph_r \text{ and } w(V) \geq \aleph_r.$$

Proof. Such a space has a basis of compact opens. To recall why, given a point and an open neighborhood, choose a compact neighborhood inside it. This compact Hausdorff totally disconnected subspace is zero-dimensional, so it contains a clopen neighborhood of the point lying inside its own interior in V . That smaller neighborhood is both open in V and compact. Thus compact opens refine every neighborhood.

Choosing one compact-open neighborhood of each point gives a cover with at most $|V|$ members. Thus $|V| \leq \aleph_n$ implies vanishing above degree n by Corollary 6.4.

For weight, choose a basis $\mathcal{B}$ of cardinality $w(V)$. Every compact open is a finite union of basis elements: cover it by basis neighborhoods contained in it and use compactness. Choosing one such finite description for each compact open gives

$$|\text{CompactOpens}(V)| \leq \max(\aleph_0, w(V)).$$

Indeed, identical finite descriptions have identical unions, so this choice is injective into the set of finite subsets of $\mathcal{B}$. For an infinite basis, that set has the same cardinality as the basis. All compact opens together cover V . Therefore $w(V) \leq \aleph_n$ also forces vanishing above degree n ; the displayed maximum handles finite or empty bases as well.

Finally, a cardinal strictly below $\aleph_r$, for positive finite r , is at most $\aleph_{r-1}$. Either $|V| < \aleph_r$ or $w(V) < \aleph_r$ would therefore imply $H^r(V; \mathcal{M}) = 0$. Taking contrapositives proves both inequalities.^[L17] $\square$

A nonzero product of r degree-one classes is a nonzero degree- r cohomology class, so these inequalities apply to every space considered in Aoki's question. The examples of Theorem 1.1 attain both bounds simultaneously. This proves the asserted optimality for cardinality and weight.

7 What the formalization checks

The proof, formalization, and exposition were developed with AI assistance. The accompanying development formalizes the spaces, their topological properties, both exact size calculations, the finite projective resolutions, the cocycle, the cup-product calculation, the nonvanishing argument, and the universal lower bounds. The assembled theorem is about

actual spaces and ordinary sheaf cohomology, rather than an auxiliary quotient defined only to encode the combinatorial obstruction.

There are three points of translation between the exposition and the code. First, the integral free sheaf on the whole space is identified with the constant integral object in Mathlib’s definition of ordinary sheaf cohomology. Second, the cup operation is implemented through the constant-module Yoneda algebra and the proved comparison of section complexes from [section 4](#). The code proves the chain-lift formula; it does not compare this construction with an independently defined Mathlib sheaf cup-product operation. Third, although the gauge choices suggest interpreting η_r as the class of a double cover, a torsor-classification equivalence is not needed and is not formalized.

The code uses Boolean bits and the field with two elements. Its recursive configuration type indexed by r has $r + 1$ coordinates, and its alternating function has r factors. The final theorem uses the same positive degree r as [theorem 1.1](#); some inductive helpers use an index one smaller.

7.1 Reproducing the verification

The project pins Lean v4.31.0 and Mathlib v4.31.0, with Mathlib commit

```
fabf563a7c95a166b8d7b6efca11c8b4dc9d911f.
```

The transitive dependency commits are recorded in the manifest. From the project’s lean directory, run

```
lake exe cache get
lake build
lake env lean AxiomAudit.lean
```

The aggregate `Aoki.lean` imports 64 local modules. The recorded build succeeds, and the axiom audit for the principal statements and bridges reports only propositional extensionality, classical choice, and quotient soundness. The sources contain no proof placeholders or additional mathematical axioms. The output is retained in [lean/verification.txt](#).

[Appendix A](#) is a source concordance, not an additional mathematical assumption: each entry identifies definitions and proved declarations used in the proof above. File links are relative to the distributed project, and printed line numbers locate the declarations even in a viewer that does not open Lean source files.

A Lean source concordance

The following entries give the precise source locations for the markers in the text. All paths are relative to the lean directory; line numbers refer to the accompanying source snapshot. Each file name is a link to that file in the distributed project. The declarations are grouped by their mathematical role so that a reader can follow one part of the argument without reading the entire development.

L1. The assembled theorem

- [Aoki/Main.lean](#), line 31:
aoki_question_2_14.
- [Aoki/Main.lean](#), line 47:
aoki_question_2_14_minimality.

- [Aoki/Main.lean](#), line 56:
aoki_example_cohomology_above.

L2. The punctured involution quotient

- [Aoki/Topology/Quotient.lean](#), line 19:
CompactProduct.
- [Aoki/Topology/Quotient.lean](#), line 138:
CompactQuotient.
- [Aoki/Topology/Quotient.lean](#), line 163:
U.
- [Aoki/Topology/Quotient.lean](#), line 73:
flip_eq_self_iff.
- [Aoki/Topology/Quotient.lean](#), line 196:
isOpenMap_quotientMap.

L3. Gauge charts and compact-open cells

- [Aoki/Sheaf/ChartProjectivity.lean](#), line 37:
iSup_chartOpen.
- [Aoki/Topology/Charts.lean](#), line 86:
sliceHomeomorph.
- [Aoki/Topology/Charts.lean](#), line 126:
chartProductHomeomorph.
- [Aoki/Topology/Intersections.lean](#), line 14:
GaugeLabel.
- [Aoki/Topology/Intersections.lean](#), line 151:
cellOpen.
- [Aoki/Topology/Intersections.lean](#), line 84:
cell_pairwiseDisjoint.
- [Aoki/Topology/Intersections.lean](#), line 92:
cells_cover_intersection.
- [Aoki/Topology/Intersections.lean](#), line 126:
isClopen_cell.
- [Aoki/Topology/Intersections.lean](#), line 130:
isCompact_cell.

L4. Cardinality and weight

- [Aoki/Topology/CompactWeight.lean](#), line 18:
weight_le_max_mk.
- [Aoki/Topology/Axis.lean](#), line 37:
isEmbedding_axis.
- [Aoki/Topology/GeometrySize.lean](#), line 52:
mk_U.
- [Aoki/Topology/GeometrySize.lean](#), line 74:
weight_U.
- [Aoki/Topology/Weight.lean](#), line 26:
weight.

L5. Represented free sheaves and projectivity

- [Aoki/Sheaf/FreeOpen.lean](#), line 33:
freeOpenHomEquiv.
- [Aoki/Sheaf/FreeOpen.lean](#), line 97:
freeOpenTopIso.
- [Aoki/Sheaf/ModuleFreeOpen.lean](#), line 31:
freeOpenModuleHomEquiv.
- [Aoki/Sheaf/ModuleFreeOpen.lean](#), line 97:
freeOpenModuleTopIso.
- [Aoki/Topology/CompactOpenCovers.lean](#), line 79:
exists_compactOpen_disjoint_refinement.
- [Aoki/Sheaf/EpiSections.lean](#), line 70:
surjective_sections_of_disjoint_refinements.
- [Aoki/Sheaf/ProjectiveOpen.lean](#), line 41:
projective_freeOpen_disjoint_iSup.
- [Aoki/Sheaf/ModuleFreeOpen.lean](#), line 184:
projective_freeOpenModule_disjoint_iSup.
- [Aoki/Sheaf/ChartProjectivity.lean](#), line 64:
projective_freeOpen_chartIntersection.
- [Aoki/Sheaf/ModuleChartProjectivity.lean](#), line 25:
projective_freeOpenModule_chartIntersection.

L6. The finite projective resolutions

- [Aoki/Cech/OrderedComplex.lean](#), line 161:
orderedComplex.
- [Aoki/Cech/OrderedComplex.lean](#), line 178:
orderedTerm_isZero.
- [Aoki/Cech/OrderedCone.lean](#), line 194:
orderedContraction_identity.
- [Aoki/Cech/AugmentedCone.lean](#), line 70:
orderedContraction_augmentation_identity.
- [Aoki/Cech/SheafResolution.lean](#), line 49:
freeOpenProjectiveResolution.
- [Aoki/Cech/ModuleSheafResolution.lean](#), line 81:
freeOpenModuleProjectiveResolution.

L7. Ordinary cohomology and upper vanishing

- [Aoki/Cech/NonzeroClass.lean](#), line 32:
quotientIntegralResolution.
- [Aoki/Cech/NonzeroClass.lean](#), line 70:
quotient_sheaf_cohomology_eq_zero_above.
- [Aoki/Sheaf/LocallyConstant.lean](#), line 112:
constantSheafSectionAddEquiv.

L8. The degree-one transition cocycle

- [Aoki/Cech/Cocycle.lean](#), line 70: `transitionOnOpen`.
- [Aoki/Cech/Cocycle.lean](#), line 155: `transitionCochain`.
- [Aoki/Cech/Cocycle.lean](#), line 167: `transitionCochain_cocycle`.
- [Aoki/Cech/NonzeroClass.lean](#), line 51: `degreeOneClass`.
- [Aoki/Cech/Cocycle.lean](#), line 119: `constantF2Sheaf`.

L9. The module-to-abelian comparison

- [Aoki/Cech/ModuleComparison.lean](#), line 127: `moduleCechHomEquiv_d`.
- [Aoki/Sheaf/ModuleConstantComparison.lean](#), line 125: `freeOpenModuleTopSectionAddEquiv_generator`.
- [Aoki/Cech/CupCohomology.lean](#), line 86: `cupCohomologyAddEquiv`.
- [Aoki/Cech/CupCohomology.lean](#), line 104: `cupCohomologyAddEquiv_extMk`.
- [Aoki/Cech/CupCohomology.lean](#), line 118: `cupProduct`.
- [Aoki/Cech/CupCohomology.lean](#), line 127: `cupPower`.

L10. The chain lift and cup powers

- [Aoki/Sheaf/LocallyConstantScalar.lean](#), line 72: `locallyConstantSMul`.
- [Aoki/Cech/CupProduct.lean](#), line 45: `cupLift`.
- [Aoki/Cech/CupProduct.lean](#), line 197: `cupLift_chain_signed`.
- [Aoki/Cech/CupProduct.lean](#), line 231: `ι_cupIterate`.
- [Aoki/Homological/ExtMkComposition.lean](#), line 116: `extMk_comp_of_lift`.
- [Aoki/Cech/CupExt.lean](#), line 62: `yonedaPower_cupExtClass`.
- [Aoki/Cech/CupComparison.lean](#), line 51: `cupIterate_comparison`.
- [Aoki/Cech/CupNonvanishing.lean](#), line 49: `cupPower_degreeOneClass_top`.

L11. Necessary properties of a boundary

- [Aoki/OnePoint.lean](#), line 64:
almostConstant_of_continuous_onePoint.
- [Aoki/Cech/BoundaryObstruction.lean](#), line 142:
invariant_deleted_section.
- [Aoki/Cech/BoundaryObstruction.lean](#), line 152:
along_deleted_section.
- [Aoki/Cech/CochainValues.lean](#), line 68:
cochainValue_differential.

L12. Uniform thinning and cardinal growth

- [Aoki/Thinning.lean](#), line 18:
mk_exceptional_first_coordinates_le.
- [Aoki/Thinning.lean](#), line 57:
exists_fresh_pair.
- [Aoki/Cardinals.lean](#), line 45:
mk_config.
- [Aoki/Cardinals.lean](#), line 62:
config_lt_next.

L13. Finite differences and nondecomposition

- [Aoki/Functions.lean](#), line 67:
delta_alternating.
- [Aoki/Functions.lean](#), line 73:
invariant_delta.
- [Aoki/AlmostConstant.lean](#), line 78:
not_flip_sum_one.
- [Aoki/Nonvanishing.lean](#), line 73:
alternating_one_not_boundary.
- [Aoki/Nonvanishing.lean](#), line 51:
boundary_delta.
- [Aoki/Nonvanishing.lean](#), line 120:
aleph_nondecomposition.

L14. Nonzero actual cup powers

- [Aoki/Cech/CochainValues.lean](#), line 140:
aleph_powerCochain_not_boundary.
- [Aoki/Cech/NonzeroClass.lean](#), line 62:
aleph_topPowerClass_ne_zero.
- [Aoki/Cech/CupNonvanishing.lean](#), line 58:
aleph_cupPower_ne_zero.
- [Aoki/Cech/CupNonvanishing.lean](#), line 66:
cupPower_next_eq_zero.

L15. Closure under transfinite extensions

- [Aoki/Homological/TransfiniteDimension.lean](#), line 24: `extOne_vanish_iff_surjective_comp`.
- [Aoki/Homological/TransfiniteDimension.lean](#), line 86: `extOne_vanish_of_transfinite`.
- [Aoki/Homological/TransfiniteDimension.lean](#), line 107: `ext_succ_vanish_of_transfinite`.
- [Aoki/Homological/TransfiniteDimension.lean](#), line 134: `hasProjectiveDimensionLE_of_transfinite`.

L16. Compact-open covers and projective dimension

- [Aoki/Sheaf/ProjectiveOpen.lean](#), line 52: `projective_freeOpen_countable_iSup`.
- [Aoki/Sheaf/FreeOpenColimit.lean](#), line 57: `freeOpenUnionCoconeIsColimit`.
- [Aoki/Sheaf/FreeOpenFiltration.lean](#), line 74: `initialFreeOpenFiltration_continuous`.
- [Aoki/Sheaf/FreeOpenExact.lean](#), line 64: `freeOpenMayerVietoris_shortExact`.
- [Aoki/Sheaf/FreeOpenExact.lean](#), line 70: `freeOpenUnionQuotientIso`.
- [Aoki/Sheaf/CardinalDimension.lean](#), line 54: `freeOpen_iSup_projectiveDimensionLE`.
- [Aoki/Sheaf/CardinalDimension.lean](#), line 111: `sheaf_cohomology_eq_zero_of_compactOpen_cover_cardinal`.

L17. Universal bounds from cardinality and weight

- [Aoki/Topology/Weight.lean](#), line 93: `exists_point_indexed_compactOpen_cover`.
- [Aoki/Topology/Weight.lean](#), line 128: `mk_compactOpens_le_weight`.
- [Aoki/Sheaf/CardinalDimension.lean](#), line 149: `sheaf_cohomology_eq_zero_of_cardinal_lt_aleph`.
- [Aoki/Sheaf/CardinalDimension.lean](#), line 161: `sheaf_cohomology_eq_zero_of_weight_lt_aleph`.

References

- [1] Ko Aoki, *On cohomology of locally profinite sets*, [arXiv:2411.05995v1](#) (2024). All theorem and question numbers quoted here refer to this version.
- [2] The mathlib community, *Mathlib*, version v4.31.0, [source at the pinned commit](#).